In [1]:
```
import sys
print(sys.executable)
```

```
C:\Users\Administrator\Documents\CMU\financial_data_analysis_1\.venv\Scripts\python.exe
```

In [2]:
```
import yfinance as yf
import pandas as pd
import statsmodels.api as sm
import numpy as np
import matplotlib.pyplot as plt
```

In [3]:
```
target = "JPM"
hedge = "BAC"

prices = yf.download([target, hedge], start="2025-01-01", end="2026-01-01")["Close"]

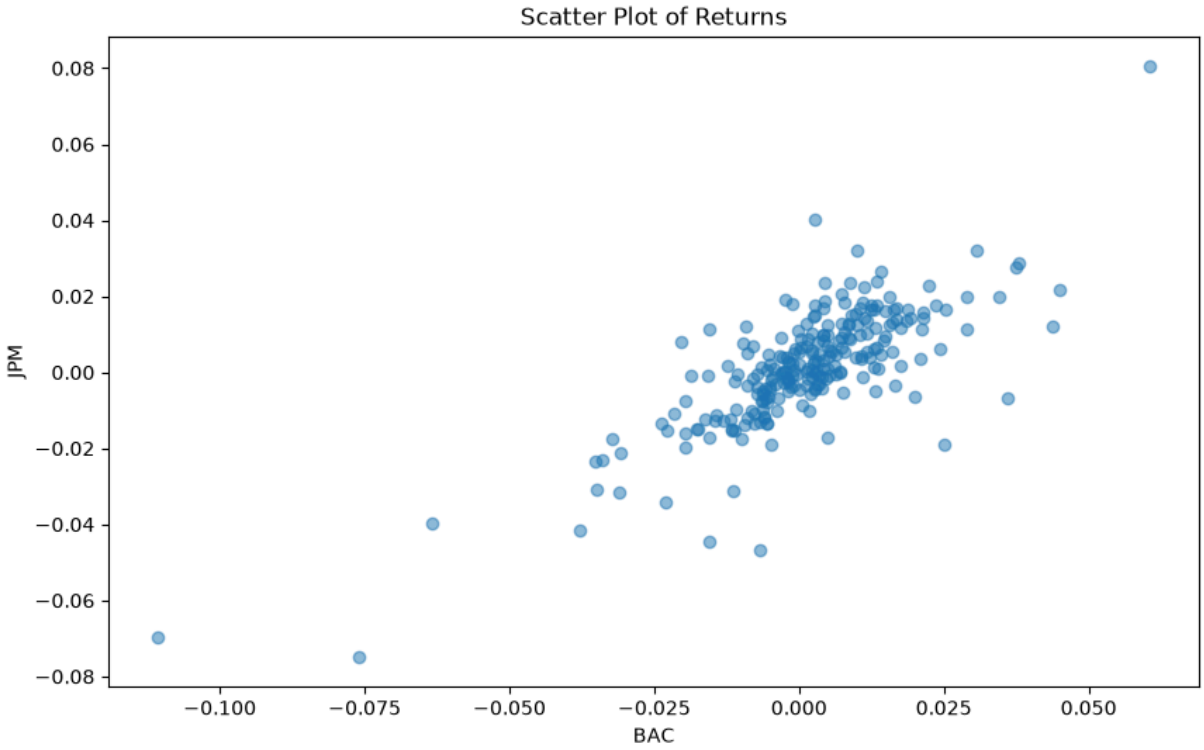
prices = prices.dropna()

returns = prices.pct_change().dropna()
```

```
[                       0%                       ]
[**********************100%**********************]  2 of 2 completed
```

In [4]:
```
RY=returns[target]
RX=returns[hedge]

plt.figure(figsize=(10, 6))
plt.scatter(RX, RY, alpha=0.5)
plt.xlabel(hedge)
plt.ylabel(target)
plt.title("Scatter Plot of Returns")
plt.show()
```



In [5]:
```
X=sm.add_constant(RX)  # Add a constant term for the intercept
model = sm.OLS(RY, X).fit()  # Fit the OLS model
```

In [6]:
```
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    JPM   R-squared:                       0.607
Model:                            OLS   Adj. R-squared:                  0.606
Method:                 Least Squares   F-statistic:                     381.9
Date:                Sat, 05 Sep 2026   Prob (F-statistic):           4.89e-52
Time:                        15:26:23   Log-Likelihood:                 798.13
No. Observations:                 249   AIC:                            -1592.
Df Residuals:                     247   BIC:                            -1585.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0006      0.001      0.942      0.347      -0.001       0.002
BAC            0.7250      0.037     19.541      0.000       0.652       0.798
==============================================================================
Omnibus:                       32.355   Durbin-Watson:                   1.942
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              141.691
Skew:                          -0.361   Prob(JB):                     1.71e-31
Kurtosis:                       6.624   Cond. No.                         59.4
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly speci
fied.
```

In [7]:
```
hedge_ratio = model.params[hedge]
print(f"Hedge Ratio (Beta): {hedge_ratio}")
```

```
Hedge Ratio (Beta): 0.724999976228322
```

In [8]:
```
CI = model.conf_int(alpha=0.05).loc[hedge]
print(f"95% Confidence Interval for Hedge Ratio: {CI.values}")
```

```
95% Confidence Interval for Hedge Ratio: [0.65192515 0.79807481]
```

In [9]:
```
hedged_returns = RY - hedge_ratio * RX

target_variance = RY.var()
hedged_variance = hedged_returns.var()
print(f"Variance of {target} Returns: {target_variance}")
print(f"Variance of Hedged Returns: {hedged_variance}")

elimation_ratio = (target_variance - hedged_variance) / target_variance
print(f"Variance Elimination Ratio: {elimation_ratio * 100:.2f}%")
```

```
Variance of JPM Returns: 0.0002460374559122898
Variance of Hedged Returns: 9.663733637824308e-05
Variance Elimination Ratio: 60.72%
```

In [10]:
```
correlation = RY.corr(RX)
print(f"Correlation between {target} and {hedge} Returns: {correlation}")
```

```
Correlation between JPM and BAC Returns: 0.7792464970117344
```